

Modelling the Optimal Distribution of Wastewater-Based Surveillance Techniques

Abhay Katoch^{1, 2}✉ and [Add Additional Authors]¹

¹Department of Biology, Queen's University at Kingston, Kingston, ON

²Existential Risk Laboratory, University of Chicago, Chicago, IL

Wastewater-based surveillance (WBS) techniques were used during the COVID-19 pandemic to track case progression, forecast hospitalizations, and identify novel variants. Previous research has shown that WBS systems may be more advantageous and cost-effective than traditional case-ascertainment techniques, and also effective at identifying potential future pandemics and endemic diseases. Other research has also shown that epidemiological models can forecast future case-loads and predict outcomes from non-pharmaceutical interventions (NPIs). However, as public health units deploy WBS systems, little work has been conducted on the optimal distribution and overall cost of WBS systems—limiting options for policy-makers.

This study evaluates the efficacy of wastewater-based surveillance (WBS) systems in predicting COVID-19 viral prevalence across various jurisdictions in the United States with several coordination strategies. We use a compartmental epidemiological model and inter-jurisdiction commuting data to assess the efficacy of WBS techniques in identifying novel cases for a SARS-CoV-2-like pandemic. We also evaluate the strengths and weaknesses of coordination across several county-level jurisdictions for public-health responses.

Our findings suggest that WBS systems can significantly reduce the lag in case ascertainment and are amenable to inter-jurisdiction coordination—potentially reducing the cost of WBS deployment. We also determine that WBS systems are likely inexpensive compared to other public health measures. We also provide avenues for future research into reducing costs associated with WBS systems. This study contributes to the ongoing discourse on public health strategies, proposing a scalable approach to early detection and management of infectious diseases.

Epidemiological Modelling | Infectious Disease | Pandemic Preparedness
Correspondence: Abhay@Katoch.ca

Introduction

The World Health Organization (WHO) has estimated that approximately 15 million individuals have died as a result of the COVID-19 pandemic—about 2.75 times higher than official estimates (1). This discrepancy has highlighted the inadequacy of current methods for case ascertainment during pandemics.

Traditional methods for case ascertainment—such as polymerase chain reaction (PCR) or lateral flow viral antigen tests—suffer from multiple problems (2). First, these methods require individuals to undergo testing, which introduces the risk of non-compliance (2, 3). Second, most individuals only undergo testing once symptomatic, potentially inducing

a lag between infection and case ascertainment (4). Third, these methods require pathogen-specific antigens or access to PCR facilities, creating a cost barrier (5).

Novel systems, such as wastewater-based surveillance (WBS), can alleviate these issues. Pathogens such as SARS-CoV-2 shed biomarkers—such as viral RNA/DNA—into wastewater at a rate proportional to the number of infected individuals in the population (6). WBS systems track the aggregate amount of viral biomarkers at a specific point in time through the sequencing of wastewater in watersheds or treatment plants using quantitative PCR (qPCR) or next-generation sequencing (NGS) systems (7).

Previous work has shown that WBS systems effectively identify viral prevalence in a population and also provide aggregate data across an entire population without testing specific individuals (7). Viral shedding may also occur before symptom onset, reducing the delay between exposure and case ascertainment for WBS systems (6, 7). Evidence from COVID-19 has shown that WBS systems can identify cases up to 14 days before clinical tests (8). WBS systems can also ascertain cases as soon as qPCR primers are developed for a sequence (7). Some have also proposed using WBS systems to identify novel pathogens before they have been detected through other means (9).

WBS systems still face multiple issues. First, it remains cost-prohibitive to continuously sequence or conduct qPCR in all possible watersheds for viral biomarkers (see [Discussion](#)). This necessitates that some jurisdictions are not covered by WBS sequencing, which introduces equity problems for communities historically under-served by clinical testing (10). In addition, no research has been conducted on whether WBS systems can predict viral prevalence outside their respective coverage areas.

While previous work has identified that WBS systems may be cost-effective (11), little work has been conducted on their overall cost and predictive performance across jurisdictions. Here, we model a viral outbreak with SARS-CoV-2-like parameters across different jurisdictions in the United States. We model the effectiveness of different coordination strategies between jurisdictions to produce an estimate of the predictive performance of WBS systems and a model of their cost-effectiveness.

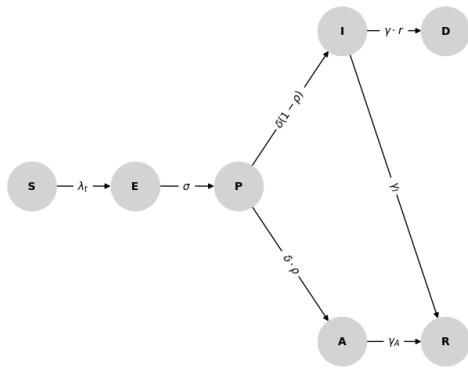


Fig. 1. Directed graph representation of epidemiological model. Each node denotes a specific state—Susceptible (S), Exposed (E), Presymptomatic (P), Infectious (I), Asymptomatic (A), Recovered (R), and Deceased (D). Edges indicate possible transition paths from each state, modelled by a binomial draw model (see S1: [Epidemiological Model](#)).

Methods

Epidemiological Model In previous work we used a compartmental model with six components: susceptible (S), exposed (E), presymptomatic (P), infectious (I), asymptomatic (A), and recovered (R) (11, 12). Here we include a seventh compartment, deceased (D) (see Fig. 1). We model these compartments stochastically using a binomial draw (see S1: [Epidemiological Model](#) and (11)). To account for the reduction in death-rates seen during the pandemic (11, 13), we implement a time-varying infection fatality rate that decreases over the length of our simulation.

Previous work has determined that SIR-based models can effectively characterize the spread of viruses (14). While most compartmental models are deterministic, we use a binomial draw equation to measure population out-flows (see S1: [Epidemiological Model](#)).

Previous research has shown that stochastic models can simulate pandemic outcomes (15). While deterministic models may be accurate for large populations, they fail to capture viral transmission dynamics at low prevalence levels (16). Evidence from the COVID-19 pandemic shows that viral transmission may follow an unpredictable pattern and never fully reach zero, as would be the case under most deterministic models (16, 17).

For simplicity, we assume that immunity does not wane over the course of the simulation, nor are removed/deceased individuals replaced. The population (N) of each jurisdiction also remains static.

Jurisdiction Information We simulate two types of jurisdictions: first, we simulate a set of 6 jurisdictions with homogeneous populations $N_i = 1,000,000$ and symmetrical mixing profiles to create a generalized case for the distribution of WBS systems. Second, we simulate jurisdictions based on the 2023 US Census data and ACS flows for Massachusetts to explore the external validity of our findings for real-world jurisdictions $N_{\text{tot}} = 7,032,933$. For more information on parameters used for jurisdictions, see S1: [Epidemiological Model](#).

We assume that certain individuals transit between each jurisdiction, contributing to overall viral spread. We created an $n \times n$ inter-jurisdiction 'mixing matrix' (β_b) to characterize the number of individuals that spread the virus across each jurisdiction. As discussed in previous work (11), we partition overall mixing into three categories: mixing at home (k_h), mixing at work/travel (k_{wt}), and other cases (k_o). Information on the proportion of inter-jurisdiction mixing was obtained from 2023 US Census ACS commuting flows. The relative proportion of mixing from k_h , k_{wt} , and k_o was informed by previous work (12). The overall mixing matrix (β_b) is the sum of mixing in each category. For more information, see S2: [Jurisdiction Information](#).

Coordination Each jurisdiction models target NPI policies (NPI) as a function of ascertained disease incidence (see Eq. 1). Jurisdictions will also modify target NPI policies based on the NPI policies of other jurisdictions, depending the degree of coordination (C).

As mentioned above, we model the coordination in NPI policies for jurisdictions with uneven WBS regimes. We create an $n \times n$ coordination matrix ($C \in [0, 1]$) that models the degree of correlation between the non-pharmaceutical intervention (NPI) policies of any two jurisdictions (see S2: [Jurisdiction Information](#)). Each jurisdiction updates coordination policies with respect to others if $C_{i,j} \neq 0$ and if $\text{NPI}_j > \text{NPI}_i$.

We modelled five types of inter-jurisdiction coordination. We chose policies that were representative of NPI coordination policies seen during the COVID-19 pandemic, as referenced by the Oxford COVID-19 Government Response Tracker (OxCGRT) (18). For more information on specific coordination policies used, see S3: [Coordination](#).

Scenarios To compare the efficacy of our models and to test each coordination strategy, we modelled five scenarios: First, to test the efficacy of NPI policies, we created a no-NPI model, in which jurisdictions do not implement NPI policies based on case-ascertainment techniques. Second, as mentioned previously, we created homo- and heterogeneous model scenarios, in which jurisdictions either have uniform populations and symmetric commuting patterns (no-het.) or have population and commuting structures based on data obtained from Massachusetts (see S3: [Scenarios](#)). Third, we also simulated the marginal effect of WBS systems, in which some jurisdictions have access to WBS, while others do not (See [Marginal Benefit of WBS](#)).

Results

A. Model Comparison Previous research has shown that viral DNA/RNA concentrations in wastewater were an accurate predictor of total case-counts and can ascertain cases more quickly and accurately than other determination methods—such as traditional 'test-and-trace' techniques. We sought to determine whether the implementation of NPI policies in accordance with wastewater surveillance would decrease projected case-counts and death rates in various long-term scenarios.

Table 1 validates previous findings that the implementation of WBS reduces case-counts, death rates, and costs associated with illness. Previous work has shown that in models where no NPIs are implemented, death rates per 100,000 individuals are higher than in real-world scenarios. We validate these findings.

For generalized, non-heterogeneous scenarios, we find large differences between no-NPI, NPI without WBS, and NPI with WBS scenarios. Under the no-NPI scenario, we find that death rates are 542 (536-548) per 100,000 individuals. We find that the addition of NPIs, without the use of WBS systems, reduces death rates to 542 (536-548) per 100,000 individuals ($F = 34.23$, $p < 0.001$). The overall cost of illness also decreases from \$66,500 (\$65,800 - \$67,200) to \$66,500 (\$65,800 - \$67,200) ($F = 34.23$, $p < 0.001$).

We also find a benefit to the addition of WBS systems; we find that, in comparison to NPI without WBS scenarios, death rates decrease to 542 (536-548) per 100,000 individuals ($F = 34.23$, $p < 0.001$). WBS systems also increase NMB from \$66,500 (\$65,800 - \$67,200) to \$66,500 (\$65,800 - \$67,200) ($F = 34.23$, $p < 0.001$).

For our heterogeneous scenarios, we find a similar decrease in death rates between no-NPI, NPI without WBS, and NPI with WBS scenarios. Death rates decrease from 542 (536-548) from the no-NPI scenarios to 542 (536-548) for NPI without WBS scenarios, and 542 (536-548) for NPI with WBS scenarios ($F = 34.23$, $p < 0.001$). NMB increases from \$66,500 (\$65,800 - \$67,200) to \$66,500 (\$65,800 - \$67,200) and \$66,500 (\$65,800 - \$67,200) between all three scenarios.

Marginal Benefit of WBS Fig. 2 shows the NMB for the marginal addition of WBS to one jurisdiction in comparison to an NPI without WBS counterfactual scenario. While WBS systems have large NMB when deployed uniformly across all jurisdiction, we find that the marginal benefit does not scale linearly with each jurisdiction added. We found that when jurisdictions do not coordinate with regards to NPI policies, the addition of WBS systems to each jurisdiction follows a roughly logarithmic pattern: the addition of WBS to initial jurisdictions has a larger effect on overall NMB than successive jurisdictions.

However, the NMB accrued for the addition of most jurisdictions remains significant overall ($F = 34.23$, $p > 0.05$). For the non-heterogeneous scenario, the addition of WBS is

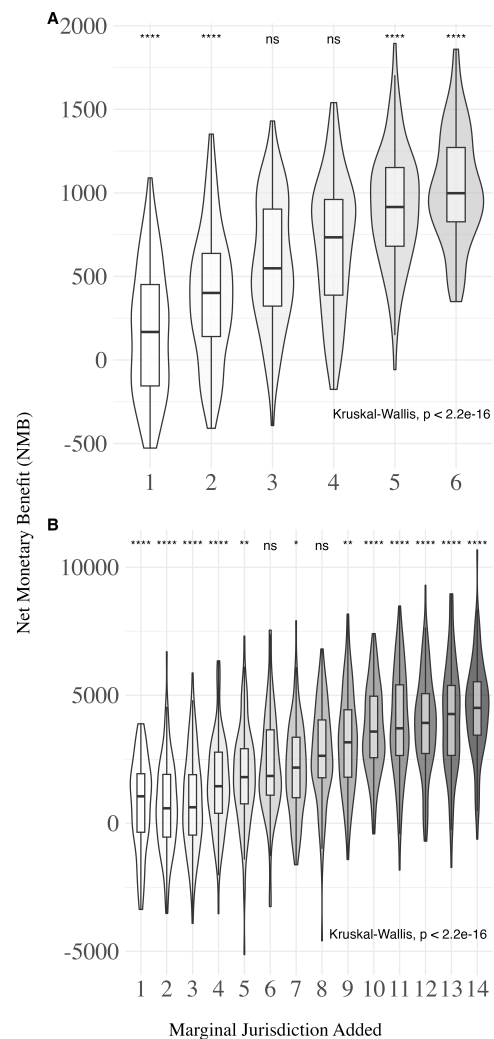


Fig. 2. Net Monetary Benefit of WBS for Homogeneous and Heterogeneous Jurisdictions. A) shows the net monetary benefit (NMB) for the marginal addition of WBS for each of the 6 homogeneous scenarios. B) shows the NMB for each jurisdiction in the heterogeneous scenario—modelled after population and commuting flows in Massachusetts. Note: Each replicate is a random pairing of each jurisdiction.

significant for all jurisdictions, except the third and fourth added ($F = 34.23$, $p > 0.05$). For the heterogeneous scenario, this benefit remains significant for all jurisdictions except the fifth and seventh. We also find that the marginal increase for the heterogeneous scenario is higher than that of the non-heterogeneous scenario—potentially reflecting differences in commuting flows between real-world jurisdictions.

Scenario	Deaths per 100,000 people	Cost of Illness	Net Monetary Benefit
No NPIs (no-het.)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	-\$35,000 (-\$35,900 - -\$34,800)
NPIs w/o WBS (no-het.)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	\$0 (\$0 - \$0)
NPIs w/ WBS (no-het.)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	-\$35,000 (-\$35,900 - -\$34,800)
No NPIs (no-het.)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	-\$35,000 (-\$35,900 - -\$34,800)
NPIs w/o WBS (het., MA)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	-\$35,000 (-\$35,900 - -\$34,800)
NPIs w/ WBS (het., MA)	542 (536-548)	\$66,500 (\$65,800 - \$67,200)	-\$35,000 (-\$35,900 - -\$34,800)

Table 1. Health and Cost Outcomes for Counter-Factual and WBS Scenarios. Counter-factual scenarios were modelled without the addition of WBS, while experimental scenarios were modelled based on either a) jurisdictions with homogeneous population sizes and symmetrical commuting patterns (no-het.) or b) heterogeneous jurisdiction sizes based on 2023 US Census commuting patterns for Massachusetts (see S4: Scenarios).

Optimal Coordination Patterns Under a no-coordination scenario, the marginal benefit of WBS is always positive: each jurisdiction gains relative information on case-ascertainment at any given time-point. However, previous research has shown that COVID-19 case-counts correlate between jurisdictions—potentially reducing the benefit of WBS for jurisdictions with large commuting populations. Therefore, we also examined the effect of multiple coordination schemes on the marginal benefit of WBS systems. For more information on the specific coordination schemes used, see [Coordination](#) and S2: [Coordination](#).

Fig. 3 shows the marginal NMB increase between coordination schemes in comparison to a no-coordination, no-WBS baseline for the heterogeneous scenario. We find that for all coordination scenarios, NMB is higher than baseline ($F = 34.23, p > 0.05$). Between coordination strategies, we also find significant differences ($F = 34.23, p > 0.05$). Coordination strategies A and B do not differ ($F = 34.23, p > 0.05$), while coordination strategy C shows lower NMB than A or B ($F = 34.23, p > 0.05$).

We also find similar outcomes for death rates between coordination strategies ($F = 34.23, p > 0.05$). No-coordination, baseline scenarios show 542 (536-548) deaths per 100,000 individuals, while coordination strategies A and B show 542 (536-548) and 542 (536-548) deaths per 100,000 individuals respectively. However, coordination strategy C shows 542 (536-548) deaths per 100,000 individuals, which is lower than either strategy A or B ($F = 34.23, p > 0.05$).

The variance in model outcomes also differs between coordination strategies ($F = 34.23, p > 0.05$). The no-coordination, WBS, baseline shows the least variance in model outcomes, while coordination A shows the largest ($F = 34.23, p > 0.05$); coordination strategy B and C show successively lower variances in model outcomes, but higher than under baseline ($F = 34.23, p > 0.05$).

Overall Cost of WBS Systems

Discussion

Conclusions

ACKNOWLEDGEMENTS

DATA AVAILABILITY

All code and data that support the findings of this study have been deposited in a GitHub repository and is available publicly. To view code and datasets used for this analysis, visit: <https://github.com/Abhay-Katoch/epi-policy>.

RIGHTS AND PERMISSIONS

This study is licensed under a Creative Commons Attribution 4.0 International License. All material in this article are included in the article's Creative Commons license, unless indicated otherwise. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>.

COMPETING FINANCIAL INTERESTS

The authors report no conflicts of interest.

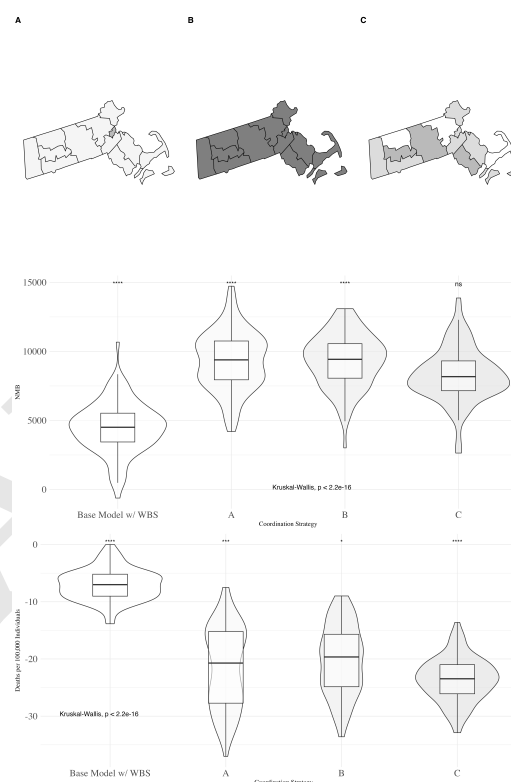


Fig. 3. Net Monetary Benefit and Deaths per 100,000 Individuals of Coordination Scenarios. 1) Net Monetary Benefit with respect to no-coordination, no-WBS, counterfactual scenario for no-coordination, WBS base-case scenario, and 3 coordination schemes. For more information on coordination schemes, see S2: [Coordination](#). 2) Deaths per 100,000 individuals for the base-case scenario and the previously-mentioned coordination schemes. 3) Distribution of WBS and coordination groups for each scenario presented.

Bibliography

1. William Msemburi, Ariel Karlinsky, Victoria Knutson, Serge Aleshin-Guendel, Somnath Chatterji, and Jon Wakefield. The WHO estimates of excess mortality associated with the COVID-19 pandemic. *Nature*, 613(7942):130–137, January 2023. ISSN 1476-4687. doi: 10.1038/s41586-022-05522-2. Publisher: Nature Publishing Group.
2. Elizabeth Alvarez, Iwona A. Bielska, Stephanie Hopkins, Ahmed A. Belal, Donna M. Goldstein, Jean Slick, Sureka Pavalagantharajah, Anna Wynfield, Shruthi Dakey, Marie-Carmel Gedeon, Edris Alam, and Katrina Bouzanis. Limitations of COVID-19 testing and case data for evidence-informed health policy and practice. *Health Research Policy and Systems*, 21(1):11, January 2023. ISSN 1478-4505. doi: 10.1186/s12961-023-00963-1.
3. Mark Embrett, S. Meaghan Sim, Hilary A. T. Caldwell, Leah Boulos, Ziwa Yu, Gina Agarwal, Rhiannon Cooper, Allyson J. Gallant AJ, Iwona A. Bielska, Jawad Chishtie, Kathryn Stone, Janet Curran, and Andrea Tricco. Barriers to and strategies to address COVID-19 testing hesitancy: a rapid scoping review. *BMC Public Health*, 22:750, April 2022. ISSN 1471-2458. doi: 10.1186/s12889-022-13127-7.
4. Chad R. Wells, Jeffrey P. Townsend, Abhishek Pandey, Seyed M. Moghadas, Gary Krieger, Burton Singer, Robert H. McDonald, Meagan C. Fitzpatrick, and Alison P. Galvani. Optimal COVID-19 quarantine and testing strategies. *Nature Communications*, 12(1):356, January 2021. ISSN 2041-1723. doi: 10.1038/s41467-020-20742-8. Publisher: Nature Publishing Group.
5. Monica Grandien. Viral diagnosis by antigen detection techniques. *Clinical and Diagnostic Virology*, 5(2):81–90, May 1996. ISSN 0928-0197. doi: 10.1016/0928-0197(96)00209-7.
6. Xi He, Eric H. Y. Lau, Peng Wu, Xilong Deng, Jian Wang, Xinxin Hao, Yiu Chung Lau, Jessica Y. Wong, Yajuan Guan, Xinghua Tan, Xiaoneng Mo, Yanqing Chen, Baolin Liao, Weilie Chen, Fengyu Hu, Qing Zhang, Mingqiu Zhong, Yanrong Wu, Lingzhai Zhao, Fuchun Zhang, Benjamin J. Cowling, Fang Li, and Gabriel M. Leung. Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nature Medicine*, 26(5):672–675, May 2020. ISSN 1546-170X. doi: 10.1038/s41591-020-0869-5. Publisher: Nature Publishing Group.
7. Michael D. Parkins, Bonita E. Lee, Nicole Acosta, Maria Bautista, Casey R. J. Hubert, Steve E. Hruddy, Kevin Frankowski, and Xiao-Li Pang. Wastewater-based surveillance as a tool for public health action: SARS-CoV-2 and beyond. *Clinical Microbiology Reviews*, 37(1):e00103–22, December 2023. doi: 10.1128/cmr.00103-22. Publisher: American Society for Microbiology.
8. Scott W. Olesen, Maxim Imakaev, and Claire Duvallet. Making waves: Defining the lead time of wastewater-based epidemiology for COVID-19. *Water Research*, 202:117433, September 2021. ISSN 0043-1354. doi: 10.1016/j.watres.2021.117433.
9. The Nucleic Acid Observatory Consortium. A Global Nucleic Acid Observatory for Biodefense and Planetary Health, August 2021. arXiv:2108.02678 [q-bio].
10. Priscilla H. Mullachery, Ran Li, Steven Melly, Jennifer Kolker, Sharrelle Barber, Ana V. Diez Roux, and Usama Bilal. Inequities in spatial accessibility to COVID-19 testing in 30 large US cities. *Social Science & Medicine (1982)*, 310:115307, October 2022. ISSN 0277-9536. doi: 10.1016/j.socscimed.2022.115307.
11. Pedro Nascimento de Lima, Sarah Karr, Jing Zhi Lim, Raffaele Vardavas, Derek Roberts, Abigail Kessler, Jalal Awan, Laura J. Faherty, and Henry H. Willis. The Value of Environmental Surveillance for Pandemic Response. Technical report, RAND Corporation, June 2024.
12. Raffaele Vardavas, Aaron Strong, Jennifer Bouey, Jonathan W. Welburn, Pedro Nascimento de Lima, Lawrence Baker, Keren Zhu, Michelle Priest, Lynn Hu, and Jeanne S. Ringel. The Health and Economic Impacts of Nonpharmaceutical Interventions to Address COVID-19: A Decision Support Tool for State and Local Policymakers. Technical report, RAND Corporation, May 2020.
13. Samiran Ghosh, Vitaly Volpert, and Malay Banerjee. An Epidemic Model with Time-Distributed Recovery and Death Rates. *Bulletin of Mathematical Biology*, 84(8):78, 2022. ISSN 0092-8240. doi: 10.1007/s11538-022-01028-0.
14. Peijue Zhang, Kairui Feng, Yuqing Gong, Jieon Lee, Sara Lomonaco, and Liang Zhao. Usage of Compartmental Models in Predicting COVID-19 Outbreaks. *The AAPS Journal*, 24(5):98, September 2022. ISSN 1550-7416. doi: 10.1208/s12248-022-00743-9.
15. J. H. Matis and T. E. Wehrly. Stochastic Models of Compartmental Systems. *Biometrics*, 35(1):199–220, 1979. ISSN 0006-341X. doi: 10.2307/2529945. Publisher: [Wiley, International Biometric Society].
16. Peter Csuppon, Emmanuel Schertzer, François Blanquart, and Florence Débarre. The stochastic dynamics of early epidemics: probability of establishment, initial growth rate, and infection cluster size at first detection. *Journal of the Royal Society, Interface*, 18(184):20210575, November 2021. ISSN 1742-5662. doi: 10.1098/rsif.2021.0575.
17. Damilola Olabode, Jordan Culp, Allison Fisher, Angela Tower, Dylan Hull-Nye, and Xueying Wang. Deterministic and stochastic models for the epidemic dynamics of COVID-19 in Wuhan, China. *Mathematical biosciences and engineering: MBE*, 18(1):950–967, January 2021. ISSN 1551-0018. doi: 10.3934/mbe.2021050.
18. Thomas Hale, Noam Angrist, Rafael Goldszmidt, Beatriz Kira, Anna Petherick, Toby Phillips, Samuel Webster, Emily Cameron-Blake, Laura Hallas, Saptarshi Majumdar, and Helen Tatlow. A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker). *Nature Human Behaviour*, 5(4):529–538, April 2021. ISSN 2397-3374. doi: 10.1038/s41562-021-01079-8. Publisher: Nature Publishing Group.

Supplementary Note 1: Epidemiological Model
Supplementary Note 2: Jurisdiction Information
Supplementary Note 3: Coordination
Supplementary Note 4: Parameter Estimation
Supplementary Note 5: Sensitivity Analyses

DRAFT